Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 1 of 10 · Question 4 part c · 4 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Number				
4	(a)	<table><tr><td>Blood group</td><td>A</td><td>B</td><td>AB</td><td>O</td></tr><tr><td>Antigen</td><td>A</td><td>B</td><td>A and B</td><td>none</td></tr></table>	all four correct = 2 1-3 correct = 1	2
	(b)	• agglutination (1) • blocks vessels (1) • causes death (1)		3
	(c)	recipient blood group A = A and O (1) recipient blood group B = B and O (1) recipient blood group AB = AB, O, A and B/allgroups (1) recipient blood group O = O (1)		4
	(d)	• 10 + 4 = 14% (1) • $\frac{14 \times 750}{100}$ (1) • = 105 (1)	full marks for correct answer	3

Total 12 marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 1 of 10 · Question 4 part c · 4 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 2 of 10 · Question 4 part a · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number				
4	(a)	<table><tr><td>Blood group</td><td>A</td><td>B</td><td>AB</td><td>O</td></tr><tr><td>Antigen</td><td>A</td><td>B</td><td>A and B</td><td>none</td></tr></table>	all four correct = 2 1-3 correct = 1	2
	(b)	• agglutination (1) • blocks vessels (1) • causes death (1)		3
	(c)	recipient blood group A = A and O (1) recipient blood group B = B and O (1) recipient blood group AB = AB, O, A and B/allgroups (1) recipient blood group O = O (1)		4
	(d)	• 10 + 4 = 14% (1) • $\frac{14 \times 750}{100}$ (1) • = 105 (1)	full marks for correct answer	3

Total 12 marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 2 of 10 · Question 4 part a · 2 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 3 of 10 · Question 4 part d · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

Number				
4	(a)	<table><tr><td>Blood group</td><td>A</td><td>B</td><td>AB</td><td>O</td></tr><tr><td>Antigen</td><td>A</td><td>B</td><td>A and B</td><td>none</td></tr></table>	all four correct = 2 1-3 correct = 1	2
	(b)	• agglutination (1) • blocks vessels (1) • causes death (1)		3
	(c)	recipient blood group A = A and O (1) recipient blood group B = B and O (1) recipient blood group AB = AB, O, A and B/allgroups (1) recipient blood group O = O (1)		4
	(d)	• 10 + 4 = 14% (1) • $\frac{14 \times 750}{100}$ (1) • = 105 (1)	full marks for correct answer	3

Total 12 marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 3 of 10 · Question 4 part d · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 4 of 10 · Question 9 part a.i · 2 marks

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

9 (a)	(i)	A, B, AB;	2 marks for all 3 correct, 1 mark for 1 or 2 correct	(2)
	(ii)	any five from • blood type O; • no antigens present; • (person) contains anti-A and anti-B antibodies; • other blood groups have A/B antigens; • immune response if mixed; • causes agglutination of red blood cells;		(5)
(b)	(i)	• (faulty allele) carried/found on female sex/X chromosome; • males affected/female carriers/occurs more in males;		(2)
	(ii)	• blood fails to clot/excessive bleeding;		(1)

Total for Question 9 = 10 marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 5 of 10 · Question 4 part b · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number				
4	(a)	<table><tr><td>Blood group</td><td>A</td><td>B</td><td>AB</td><td>O</td></tr><tr><td>Antigen</td><td>A</td><td>B</td><td>A and B</td><td>none</td></tr></table>	all four correct = 2 1-3 correct = 1	2
	(b)	• agglutination (1) • blocks vessels (1) • causes death (1)		3
	(c)	recipient blood group A = A and O (1) recipient blood group B = B and O (1) recipient blood group AB = AB, O, A and B/allgroups (1) recipient blood group O = O (1)		4
	(d)	• 10 + 4 = 14% (1) • $\frac{14 \times 750}{100}$ (1) • = 105 (1)	full marks for correct answer	3

Total 12 marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 5 of 10 · Question 4 part b · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 6 of 10 · Question 2 part a.iii · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
2	(a) (i)	• genetic material made of RNA (1) • no DNA (1)		2
	(ii)	<table><tr><th>Disease</th><th>Blood Tested (✓)</th></tr><tr><td>anaemia</td><td></td></tr><tr><td>cystic fibrosis</td><td></td></tr><tr><td>gonorrhoea</td><td>✓</td></tr><tr><td>HIV</td><td>✓</td></tr><tr><td>scurvy</td><td></td></tr></table>	-1 for each extra tick	2
	(iii)	• donor/blood transfused into person with Ebola (1) • needs to be compatible/matched/same group/not rejected (1) • otherwise agglutination/clumping occurs (1)	R clotting	3
	(iv)	• cause blood cells to burst/cells are damaged/destroyed/lose structure (1) • results in loss of function (1)		2
	(v)	• antibodies in donated blood (1) • can destroy virus in infected person (1)		2
	(b)	• uracil present, not thymine (1) • guanine would pair with cytosine/adenine with uracil (1) • percentage of G and C/A and U would be the same (1)		3

Total for question = 14 marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 6 of 10 · Question 2 part a.iii · 3 marks

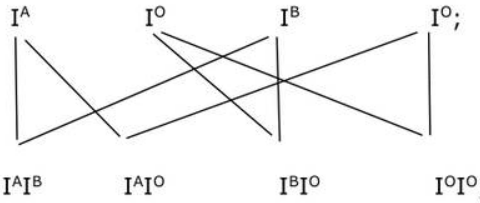
2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 7 of 10 · Question 4 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 1 of 7 · find part a.ii in this mark scheme extract

<table>
<tr>
<td>4 (a) (i)</td>
<td>
Africa
<table>
<tr><th>Blood group</th><th>Percentage of blood group</th></tr>
<tr><td>O</td><td>68</td></tr>
<tr><td>A</td><td>17/17.5</td></tr>
<tr><td>B</td><td>10;</td></tr>
<tr><td>AB</td><td>5</td></tr>
</table>
Europe
<table>
<tr><th>Blood group</th><th>Percentage of blood group</th></tr>
<tr><td>O</td><td>45</td></tr>
<tr><td>A</td><td>40</td></tr>
<tr><td>B</td><td>12</td></tr>
<tr><td>AB</td><td>3;</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
<tr>
<td>(ii)</td>
<td>
• reference to antigens on red blood cells (group B);
• antibodies produced;
• causing agglutination/clumping (of red blood cells);
</td>
<td></td>
<td>3</td>
</tr>
<tr>
<td>(iii)</td>
<td>
I^AI^O I^BI^O;


I^AI^B I^AI^O I^BI^O I^OI^O;
</td>
<td>Full marks for a Punnett square if parent genotypes are given.</td>
<td>3</td>
</tr>
<tr>
<td>(iv)</td>
<td>1 in 4:/25%/¼/0.25;</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td>(b)</td>
<td>
• <u>deoxygenated</u> blood moves into the left side/ventricle of the heart;
• (deoxygenated blood) mixes with <u>oxygenated</u> blood;
• less oxygen travelling around the body/to body/cells/in bloodstream;
• breathes faster to get more oxygen into the body;
</td>
<td></td>
<td>4

Total 13</td>
</tr>
</table>

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 7 of 10 · Question 4 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 2 of 7 · find part a.ii in this mark scheme extract

	Answer	Notes	Marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 7 of 10 · Question 4 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 3 of 7 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks
6	Any six from the following: • remove plasmid from bacteria; • locate the human insulin gene • cut/remove insulin gene (from human); • cut open the plasmid; • using restriction enzymes; • add the human insulin gene to the plasmid; • use of ligase; • ref recombinant DNA; • return the plasmid to a bacterial cell;	Suitably drawn **and** annotated diagrams can score full marks.	6
			Total 6

Question number	Answer	Notes	Marks
7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 7 of 10 · Question 4 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 4 of 7 · find part a.ii in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 7 of 10 · Question 4 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 5 of 7 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks
8 (a)	**Advantages embryonic stem cells:** • generate most cell types; • can maintain and grow for long lengths of time in culture; • useful for drug development studies/Parkinson's disease; **Disadvantages of embryonic stem cells:** • can cause tumours/cancer; • many embryos destroyed; • some people think it is unethical/killing/murder; • possibility of rejection if used in transplants **Advantages of adult stem cells:** • can be reprogrammed; • less likely rejected if used in transplants; • taken from the patient itself; **Disadvantages of adult stem cells:** • limited ability to differentiate/develop into different types of cell; • cannot be grown for long periods of time in culture; • usually a very small number found in body tissues; • can only grow small numbers;	Three marks from embryonic stem cell section and three marks from adult stem cell section Allow a valid named ethic	6
			Total 6

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 7 of 10 · Question 4 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 6 of 7 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks
9 (a)	• beta blockers; • block/bind to receptors; • which prevents the release of adrenaline/noradrenaline; • heart rate decreases; OR • ACE inhibitors; • reduce the activity of enzymes; • less angiotensin; • less narrowing of blood vessels;		4
(b)(i)	any two from • how much pressure blood is exerting; • against artery walls; • when ventricles contract/pump;		2
(ii)	• pressure in the arteries; • when the heart/ventricles at rest/relaxed/between beats;		2
			Total 8

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 7 of 10 · Question 4 part a.ii · 3 marks

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Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 8 of 10 · Question 9 part a.ii · 5 marks

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

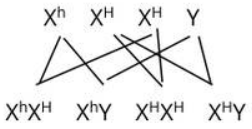
9 (a)	(i)	A, B, AB;	2 marks for all 3 correct, 1 mark for 1 or 2 correct	(2)
	(ii)	any five from • blood type O; • no antigens present; • (person) contains anti-A and anti-B antibodies; • other blood groups have A/B antigens; • immune response if mixed; • causes agglutination of red blood cells;		(5)
(b)	(i)	• (faulty allele) carried/found on female sex/X chromosome; • males affected/female carriers/occurs more in males;		(2)
	(ii)	• blood fails to clot/excessive bleeding;		(1)

Total for Question 9 = 10 marks

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 9 of 10 · Question 9 part b.i · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

9	(a)	(i)	A (the inheritance of two different alleles, both of which are expressed) B is incorrect as one allele is not expressed in codominant inheritance C is incorrect as the ABO inheritance does not involve the inheritance of multiple alleles D is incorrect as ABO inheritances does not involve the inheritance of multiple alleles		1
		(ii)	I^AI^A and I^AI^o		1
	(b)	(i)	• blood group O contains no antigens; • antigens removed/digested (by enzyme); • carbohydrase digest carbohydrates; • protease digest proteins;		Max 3
		(ii)	reduces need to find donor/cheaper/blood readily available/less chance of transfusing incorrect blood group/less risk of disease/infection;		1
	(c)	(i)	parent genotypes X^hX^H and X^HY; gametes and linkage X^h X^H X^H Y  ;	One mark for parent genotypes and one mark for gametes **and** linkage	1 1
		(ii)	1 in 4/25%/quarter/1/4 /1:3/0.25;		1

Total question 9 = 9 mark

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 9 of 10 · Question 9 part b.i · 3 marks

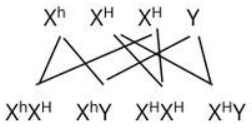
2019 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 10 of 10 · Question 9 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

9	(a)	(i)	A (the inheritance of two different alleles, both of which are expressed) B is incorrect as one allele is not expressed in codominant inheritance C is incorrect as the ABO inheritance does not involve the inheritance of multiple alleles D is incorrect as ABO inheritances does not involve the inheritance of multiple alleles		1
		(ii)	I^AI^A and I^AI^o		1
	(b)	(i)	• blood group O contains no antigens; • antigens removed/digested (by enzyme); • carbohydrase digest carbohydrates; • protease digest proteins;		Max 3
		(ii)	reduces need to find donor/cheaper/blood readily available/less chance of transfusing incorrect blood group/less risk of disease/infection;		1
	(c)	(i)	parent genotypes X^hX^H and X^HY; gametes and linkage X^h X^H X^H Y  ;	One mark for parent genotypes and one mark for gametes **and** linkage	1 1
		(ii)	1 in 4/25%/quarter/1/4 /1:3/0.25;		1

Total question 9 = 9 mark

Type MS · Use ABO antigens and antibodies to choose donors and explain clumping

Answer 10 of 10 · Question 9 part b.ii · 1 mark

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